Capital Asset Pricing Model

Professor Philip Howard
howardpd@wfu.edu

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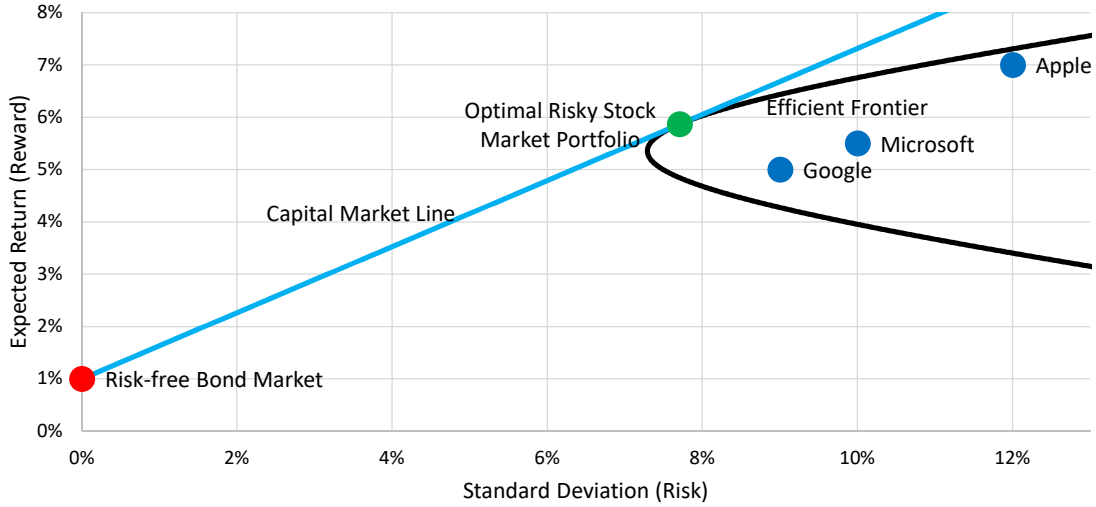
Capital Market Line



Capital Market Line

How do we price an asset on the CML?

- Plot R_i vs. σ_i
- y-intercept $= R_f$
- Slope $= \frac{R_M - R_f}{\sigma_M}$
  - Sharpe ratio of Market

$$R_i = R_f + \left(\frac{R_M - R_f}{\sigma_M}\right)\sigma_i$$

CAPM Derivation

How do we price an asset not on the CML?

$$\begin{aligned} R_i &= R_f + \left(\frac{R_M - R_f}{\sigma_M}\right) \cdot \sigma_i \cdot \rho_{iM} \\ &= R_f + \left(\frac{R_M - R_f}{\sigma_M}\right) \cdot \sigma_i \cdot \frac{\sigma_{iM}}{\sigma_i \cdot \sigma_M} \\ &= R_f + \frac{\sigma_{iM}}{\sigma_M^2}\left(R_M - R_f\right) \\ &= R_f + \beta_i\left(R_M - R_f\right) \end{aligned}$$

- σ_i and σ_M are the standard deviation of asset i and the market
- $\rho_{iM} = \frac{\sigma_{iM}}{\sigma_i \sigma_M}$: Correlation between asset i and market
- σ_{iM}: Covariance between asset i and market
- $\beta_i = \frac{\sigma_{iM}}{\sigma_M^2}$: Asset i's beta

CAPM

$$R_i = R_f + \beta_i\left(R_M - R_f\right)$$

- R_i: Expected return on asset i
- β_i: Asset i's beta
- R_M: Expected return on market
- R_f: Risk-free rate

CAPM Interpretation

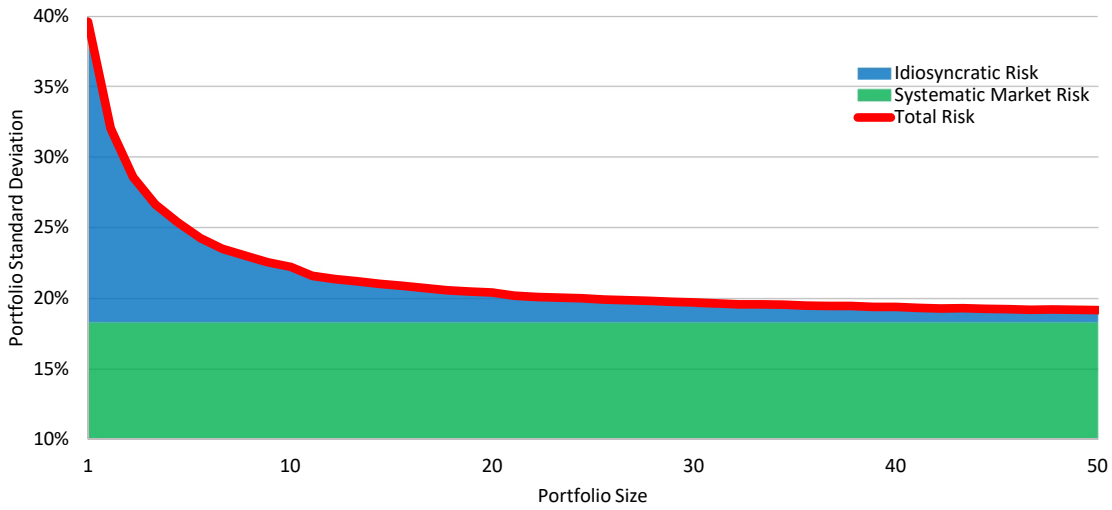
$$R_i - R_f = \beta_i \times \left(R_M - R_f\right)$$

- $R_i - R_f$: Risk premium on asset i
  - The price of asset i's risk
- β_i: Comovement of asset i with the market
  - Quantity of market risk
- $R_M - R_f$: Market risk premium
  - Price of market risk

- Interpretation

$$\text{Price of Asset } i\text{'s Risk} = \text{Quantity of Market Risk} \times \text{Price of Market Risk}$$

Diversification & Risk Compensation



Diversification & Risk Compensation

- An asset's total risk can be decomposed into two sources

$$\text{Total Risk} = \text{Systematic Risk} + \text{Idiosyncratic Risk}$$

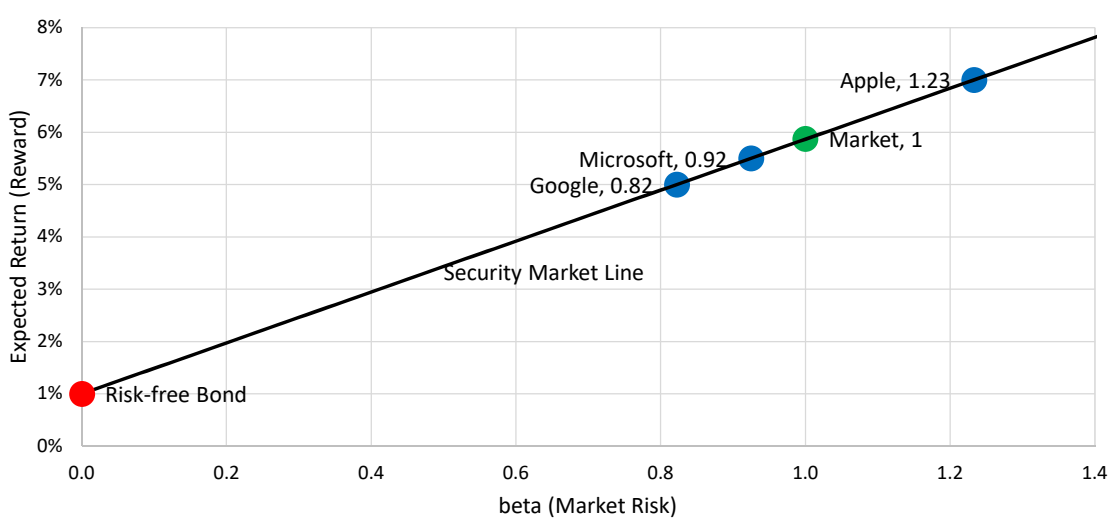
- The CAPM says that only systematic risk is priced, i.e. investors only receive a risk premium for an asset's exposure to market risk, measured by β

Security Market Line

- Plot R_i vs. β_i
- y-intercept $= R_f$
- Slope $= R_M - R_f$
  - Market risk premium

$$R_i = R_f + \beta_i\left(R_M - R_f\right)$$

Security Market Line



Estimating CAPM via Regression Analysis

- CAPM: $R_i - R_f = \beta_i \times \left(R_M - R_f\right)$
- Regression: $y = \alpha + \beta \cdot x$
  - Seek to explain y with x
  - Regress y onto x
  - Estimate y-intercept α and slope β
- Apply regression to CAPM
  - $y \equiv R_i - R_f$
  - $x \equiv R_M - R_f$
  - Seek to explain asset i's excess returns with market excess returns
  - Regress asset i's excess returns onto market excess returns

Estimating CAPM via Regression Analysis

- Regression

$$\hat{y} = \hat{\alpha} + \hat{\beta} \cdot x$$

  - Given estimated coefficients $\hat{\alpha}$ and $\hat{\beta}$, model predicts $\hat{y}$ if x occurs
- Mathematically, we have

$$\bar{y} = \hat{\alpha} + \hat{\beta} \cdot \bar{x}$$

  - $\bar{y}$ and $\bar{x}$ are average of y and x
- CAPM

$$E[R_i] - R_f = \hat{\alpha}_i + \hat{\beta}_i \cdot \left(E[R_M] - R_f\right)$$

$$RP_i = \hat{\alpha}_i + \hat{\beta}_i \cdot RP_M$$

  - The average excess return is the risk premium: $RP = E[R] - R_f$

Risk Decomposition

- To price asset i, we can decompose asset i's risk premium into
$$RP_i = \hat{\alpha}_i + \hat{\beta}_i \cdot RP_M$$
- RP_i: Risk premium on asset i
  - The price of asset i's risk
- β_i: Comovement of asset i with the market
  - Quantity of market risk
- RP_M: Market risk premium
  - Price of market risk

- Interpretation

$$\text{Price of Asset } i\text{'s Risk} = \text{Quantity of Market Risk} \times \text{Price of Market Risk}$$

CAPM Estimation Interpretations

- Estimate α, β
$$R_i - R_f = \alpha + \beta \cdot (R_M - R_F)$$

Important estimates to consider:

- Finance theory says $\alpha = 0$. If it is not, how do we interpret this?
- If $\beta = 0$, market risk factor is unimportant
- If $\beta > 1$, asset is riskier than market and commands higher risk premium

Apple's CAPM Estimation

Apple's CAPM Interpretation

- Market's monthly risk premium: $R_M - R_f = 0.62\%$
- Apple's monthly risk premium: $R_A - R_f = 1.84\%$
- Apple's beta: $\beta_A = 1.42$

- Risk premium decomposition:

$$R_A - R_f = \beta_A \times (R_M - R_f)$$
$$1.84\% \approx 1.42 \times 0.62\%$$

- Annualized:

$$22\% \approx 1.42 \times 7.4\%$$

CAPM Estimation Interpretations

- Estimate α, β

$$R_i - R_f = \alpha + \beta \cdot (R_M - R_F)$$

Important estimates to consider:

- Finance theory says $\alpha = 0$. If it is not, how do we interpret this?
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Inference

Is the CAPM model good statistically?

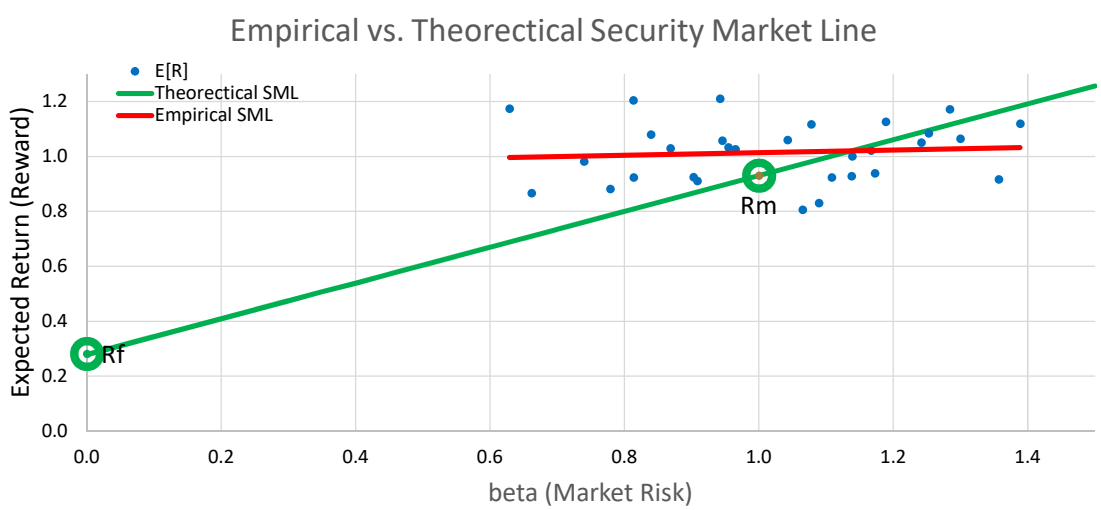
1. Hypothesis Test: $\beta = 0$
   - $\beta \neq 0 \Rightarrow$ Market risk factor is important
2. ANOVA: r^2
   - $r^2 \ll 100\% \Rightarrow$ Market is not the only important factor

Finance Theory

$\alpha = 0$

- If $\alpha > 0$, two interpretations
  1. Skill
  2. Missing risk factors
  - Empirically we cannot find persistent $\alpha > 0$

$\beta > 1$

- Asset is riskier than market and commands higher risk premium

Empirical Security Market Line